
L1 primary
L2 generator-disjoint
L3 human-authored (2000s)
Contemporary (Nazario 2015–2025 + real legitimate)
(a) Label-free prevalence estimates
CC
ACC, t = 0.5
ACC, 95%-recall t
EM
median ≤ 0.01 (at floor)
true π = 0.1%
true π = 1%
true π = 5%
Estimated / true prevalence
(median over models × partitions)
1000
100
10
1
0.1
≤0.01
L1
L2
L3
Cont.
(b) Held-out rule validation (all models)
Three-band rule
(as stated)
Always validation
cost-min. t
Decile map
learned on L1
Always validation
95%-recall t
Always t = 0.5
10%
100%
1000%
Mean excess cost
over best rule
0
0.5
1
Agreement with
best rule
(c) Sensitivity to the score distribution
word TF-IDF and MiniLM, augmented; all partitions
validation and target perturbed (a detector with another score distribution)
target only perturbed (score shift at deployment)
πr purity
1.00
0.95
0.90
0.85
Three-band rule:
mean excess cost
300%
100%
30%
10%
none
0.5
2
3
−2
−1
+1
+2
0.5
1
2
baseline
temperature
logit shift
logit noise (SD)
0.5
2
3
−2
−1
+1
+2
0.5
1
2
temperature
logit shift
logit noise (SD)